

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (3 pts) A function can have two different horizontal asymptotes.

☒ True☐ False

Why? $\tan^{-1} x$ has $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$.

/3

2. (3 pts) If $g(-2) = -5$ and $g(4) = 7$, then $g(c) = 0$ for some c in $(-2, 4)$.

☒ True☐ False

Why? **The Intermediate Value Theorem requires g to be continuous on $[-2, 4]$, and nothing here says it is. A jump can step over 0 without ever taking the value.**

/3

3. (4 pts) Evaluate the limit. **Show the step where the sign of x matters.**

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 3}}{x + 5}$$

-2

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 3}}{x + 5} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{4x^2 + 3}}{x}}{\frac{x+5}{x}}$$

divide top and bottom by x

$$= \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{4x^2 + 3}}{-\sqrt{x^2}}}{1 + \frac{5}{x}}$$

here: $\frac{1}{x} = \frac{-1}{\sqrt{x^2}}$, because $x < 0$

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{4x^2 + 3}{x^2}}}{1 + \frac{5}{x}}$$

one root over another is the root of the quotient

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{4 + \frac{3}{x^2}}}{1 + \frac{5}{x}}$$

divide inside

$$= \frac{-\sqrt{4 + 0}}{1 + 0} = -2$$

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = \lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0$$

/4

4. (5 pts) Let $f(x) = x^2 + 2x$.

(a) Find $f'(3)$ **from the definition**. Keep lim on every line until you actually take it.

$$f'(3) = 8$$

$$\begin{aligned} f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{[(3+h)^2 + 2(3+h)] - 15}{h} \\ &= \lim_{h \rightarrow 0} \frac{9 + 6h + h^2 + 6 + 2h - 15}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 8h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(h+8)}{h} = \lim_{h \rightarrow 0} (h+8) = 8 \end{aligned}$$

(b) Write the equation of the tangent line to $y = f(x)$ at $x = 3$.

$$y = 8x - 9 \quad (\text{slope 8 through } (3, 15))$$

/5

5. (3 pts) Show that the equation $x^3 + x = 1$ has a solution between 0 and 1. **Name the theorem you use, and check its hypothesis.**

/3

Let $f(x) = x^3 + x - 1$, so a solution of $x^3 + x = 1$ is a zero of f . f is a polynomial, so it is continuous on $[0, 1]$ — that is the hypothesis. Then

$$f(0) = -1 < 0 \quad \text{and} \quad f(1) = 1 > 0$$

so 0 lies between $f(0)$ and $f(1)$. By the **Intermediate Value Theorem** there is a c in $(0, 1)$ with $f(c) = 0$, and that c solves the equation.

6. (2 pts) Write the equation of the horizontal asymptote of $f(x) = \frac{3x^2 - 1}{x^2 + 4}$.

$$y = 3$$

/2

Dividing by x^2 gives $\frac{3-1/x^2}{1+4/x^2} \rightarrow \frac{3}{1}$ as $x \rightarrow \pm\infty$, so the one horizontal asymptote is $y = 3$.